Mathematics 3A03 Real Analysis I

ASSIGNMENT 4 solutions

Topic: Function Sequences

Participation deadline: **See course website**

The meaning of the participation deadline is that you must answer the multiple choice questions on childsmath before that deadline in order to receive participation credit for the assignment. The childsmath poll that you need to fill in for participation credit will be activated immediately after the last class before the above deadline.

Assignments in this course are graded only on the basis of participation, which you fulfill by answering the multiple choice questions on childsmath. You will get the same credit for any question that you answer, regardless of what your answer is. However, please answer the questions honestly so we obtain accurate statistics on how the class is doing.

You are encouraged to submit full written solutions on crowdmark. If you do so, you will not be graded on your work, but you will receive feedback that will hopefully help you to improve your mathematical skills and to prepare for the midterm test and the final exam.

There is no strict deadline for submitting written work on crowdmark for feedback, but please try to submit your solutions within a few days of the participation deadline so that the TA's work is spread out over the term. If you do not submit your solutions within a few days of the participation deadline then it may not be feasible for the TA to provide feedback via crowdmark. However, you can always ask for help with any problem during office hours with the TA or instructor.

You are encouraged to discuss and work on the problems jointly with your classmates, but remember that you will be working alone on the test and exam. You should attempt to solve the problems on your own before brainstorming with classmates, looking online, or asking the TA or instructor for help.

A full solution means either a proof or disproof of each statement that you are asked to consider when selecting your multiple choice answers.

Full solutions to the problems will be posted by the instructor. You should read the solutions only after doing your best to solve the problems, but do make sure to read the instructor's solutions carefully and ensure you understand them. If you notice any errors in the solutions, please report them to the instructor by e-mail.

Enjoy working on these problems!

– David Earn

1. Considering $f_n(x) = \sqrt[n]{x}$ on $[0,1]$, which of the following statements are true?

   ☐ $\{f_n\}$ does not converge;

   ■ $\{f_n\}$ converges pointwise;

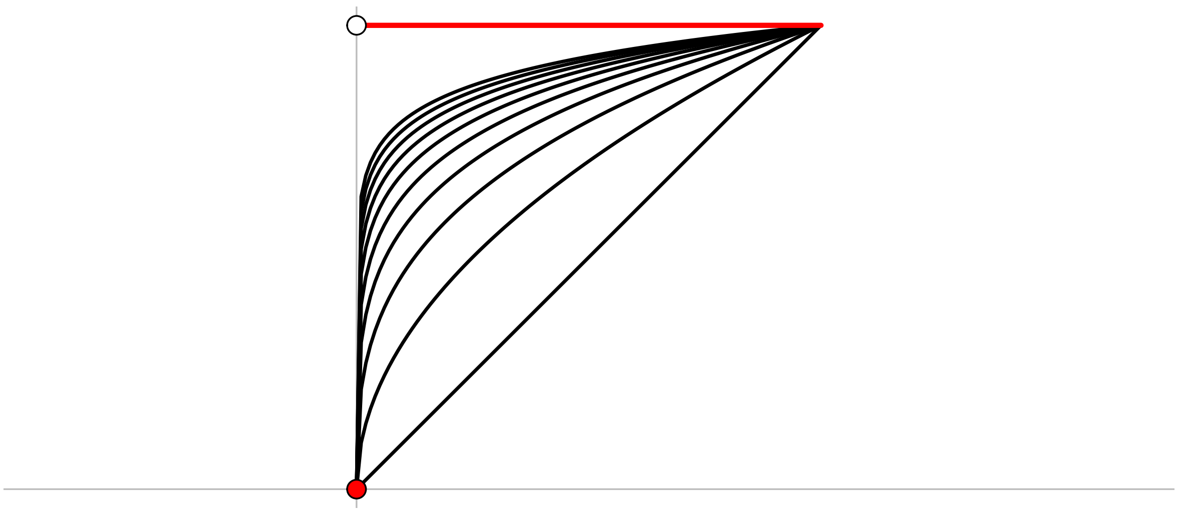
   ☐ $\{f_n\}$ converges uniformly;

   If the sequence converges pointwise, determine the pointwise limit on the indicated interval.

   The pointwise limit is

   $$f(x) = \lim_{n\to\infty} \sqrt[n]{x} = \begin{cases} 0 & x = 0, \\ 1 & 0 < x \leq 1, \end{cases}$$

   which is not continuous. Therefore, the convergence is not uniform (if it were then $f(x)$ would be continuous). In the graph below, f_n is shown in black for $n = 1, \dots, 10$ and f is shown in red.



2. Considering $f_n(x) = \begin{cases} 0, & x \leq n, \\ x - n & x \geq n, \end{cases}$ on $[a,b]$ and on $\mathbb{R}$, which of the following statements are true?

   ☐ $\{f_n\}$ does not converge;

   ■ $\{f_n\}$ converges pointwise on $[a,b]$;

   ■ $\{f_n\}$ converges pointwise on $\mathbb{R}$;

   ■ $\{f_n\}$ converges uniformly on $[a,b]$;

   ☐ $\{f_n\}$ converges uniformly on $\mathbb{R}$.

   If the sequence converges pointwise, determine the pointwise limit on the indicated interval.

The pointwise limit is $f(x) = 0$ for all $x \in \mathbb{R}$. On any finite interval $[a, b]$ this convergence is uniform. Note that for such a finite interval, we can find $N \in \mathbb{N}$ such that for all $n \geq N$, $f_n(x) = 0$ for all $x \in [a, b]$. However, the convergence is not uniform on $\mathbb{R}$ since each f_n is unbounded on $\mathbb{R}$, yet $f(x)$ is bounded. ☐

3. Considering $f_n(x) = \dfrac{e^x}{x^n}$, on $(1, \infty)$, which of the following statements are true?

   ☐ $\{f_n\}$ does not converge;

   ■ $\{f_n\}$ converges pointwise;

   ☐ $\{f_n\}$ converges uniformly;

   If the sequence converges pointwise, determine the pointwise limit on the indicated interval.

   Note that the numerator does not depend on n. Consequently, for any $x > 1$ we have the pointwise limit

$$f(x) = \lim_{n\to\infty} \frac{e^x}{x^n} = e^x \lim_{n\to\infty} \frac{1}{x^n} = 0\,.$$

   On the other hand, for any given $n \in \mathbb{N}$ we have

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{e^x}{x^n} = \infty\,,$$

   so every f_n is unbounded. So, as in the previous question, the convergence is not uniform. ☐

4. Consider the series

$$\sum_{n=1}^{\infty} \frac{x}{n(1 + nx^2)}\,.$$

   Which of the following statements are true about this series?

   ☐ does not converge for any $x \in \mathbb{R}$;

   ☐ converges pointwise on a non-empty set but not on all of $\mathbb{R}$;

   ■ converges pointwise on $\mathbb{R}$;

   ☐ converges uniformly on a non-empty set but not on all of $\mathbb{R}$;

   ■ converges uniformly on $\mathbb{R}$.

   Proof. Write

$$f_n(x) = \frac{x}{n(1 + nx^2)}\,.$$

   Then the series in question is $\sum_{n=1}^{\infty} f_n(x)$. Note that f_n is twice (in fact, infinitely) differentiable and

$$f_n'(x) = \frac{1 - nx^2}{n(1 + nx^2)}\,, \qquad f_n''(x) = -2x\frac{3 - nx^2}{(1 + nx^2)^3}\,.$$

   Thus,

$$f_n'(x) = 0 \iff x = \pm\frac{1}{\sqrt{n}}\,, \qquad f_n''(1/\sqrt{n}) < 0\,, \qquad f_n''(-1/\sqrt{n}) > 0\,,$$

so $x = 1/\sqrt{n}$ is the global maximum point and $x = -1/\sqrt{n}$ is the global minimum point of $f_n(x)$; moreover,

$$f_n\Big(\pm\frac{1}{\sqrt{n}}\Big) = \pm\frac{1}{2n^{3/2}}\,.$$

Consequently,

$$|f_n(x)| \le \frac{1}{2n^{3/2}} \qquad \forall x \in \mathbb{R}.$$

Therefore, let

$$M_n = \frac{1}{2n^{3/2}}\,,$$

and note that $\sum_{n=1}^{\infty} M_n$ converges (by the integral test). By the Weierstrass M-test, $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on $\mathbb{R}$. □

Additional practice problems

5. Consider the sequence of functions $\{f_n\}$, where

$$f_n(x) = \frac{1}{n(1+nx^2)}\,, \qquad x \in \mathbb{R}.$$

(a) For which $x \in \mathbb{R}$ does the series of functions $\displaystyle\sum_{n=1}^{\infty} f_n(x)$ converge pointwise?

For $x = 0$ we have $\sum_{n=1}^{\infty} f_n(0) = \sum_{n=1}^{\infty} \frac{1}{n}$, which diverges. For any $x \neq 0$, we have

$$0 < \sum_{n=1}^{\infty} \frac{1}{n(1+nx^2)} < \sum_{n=1}^{\infty} \frac{1}{n(nx^2)} = \frac{1}{x^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\,,$$

which converges, so $\sum_{n=1}^{\infty} f_n(x)$ converges pointwise (by the comparison test) for all $x \neq 0$. □

(b) For which $a, b \in \mathbb{R}$ $(a < b)$ does the series of functions $\displaystyle\sum_{n=1}^{\infty} f_n$ converge uniformly on $[a, b]$ to a continuous function?

From part (a) we know that any interval $[a, b]$ on which the series of functions converges uniformly cannot include $x = 0$. Hence either $a > 0$ or $b < 0$. Suppose $a > 0$. Then let

$$M_n = \frac{1}{a^2n^2}\,,$$

and note that $\sum_{n=1}^{\infty} M_n$ converges (*e.g.*, by the ratio test). Hence, by the Weierstrass M-test, $\sum_{n=1}^{\infty} f_n$ converges uniformly on $[a, b]$ to a function f, and since each f_n is continuous the uniform limit is also continuous. A similar argument establishes the result for $b < 0$, so the series converges uniformly to a continuous function on any closed interval not containing 0. □

(c) For which $a, b \in \mathbb{R}$ $(a < b)$ does the series of functions $\displaystyle\sum_{n=1}^{\infty} f_n$ converge uniformly on $[a, b]$ to a differentiable function f? For such a, b, is f' necessarily the uniform limit of $\displaystyle\sum_{n=1}^{\infty} f_n'$?

Again, since the series does not converge pointwise at $x = 0$, we must restrict attention to $a > 0$ or $b < 0$. Again, suppose $a > 0$. To apply the theorem on differentiability and uniform convergence, note that

- $f_n'(x) = -\dfrac{2x}{(1+nx^2)^2}$, so each f_n' is continuous on $[a,b]$ (in fact on $\mathbb{R}$).
- For $x \geq a > 0$,

  $$|f_n'(x)| = \frac{2\,|x|}{(1+nx^2)^2} < \frac{2\,|x|}{(nx^2)^2} = \frac{2\,|x|}{n^2x^4} = \frac{2}{|x|^3} \cdot \frac{1}{n^2} \leq \frac{2}{a^3} \cdot \frac{1}{n^2}\,.$$

  Hence, taking $M_n = 2/(a^3n^2)$, the Weierstrass M-test implies that $\sum_{n=1}^{\infty} f_n'$ converges uniformly on $[a,b]$.
- We already know from part (b) that $\sum_{n=1}^{\infty} f_n$ converges pointwise to a continuous function f.

Consequently, if $a > 0$ then f is differentiable on $[a,b]$ and $f' = \sum_{n=1}^{\infty} f_n'$ (*i.e.*, f' is the uniform limit of $\sum_{n=1}^{\infty} f_n'$). In addition, the argument is similar if $b < 0$, so the result is true for any closed interval not containing 0. $\square$

(d) Rather than closed, finite intervals $[a,b]$, consider infinite open intervals (a,∞). Answer parts (b) and (c) again after revising them to read "For which $a \in \mathbb{R}$ does the series... converge uniformly on (a,∞) to...".

The Weierstrass M-test relates to any domain $D \subseteq \mathbb{R}$. It is not necessary that D be a closed interval, or indeed any sort of interval. The arguments in parts (b) and (c) apply equally well to (a,∞) for $a > 0$ or $(-\infty,b)$ for $b < 0$. Thus, the answer to the question posed is "any $a > 0$". $\square$

6. Prove or disprove: If $f(x) = \displaystyle\sum_{n=1}^{\infty} a_nx^n$ is an even function, then $a_n = 0$ for n odd, and if f is an odd function, then $a_n = 0$ for n even.

Proof. Given the power series expansion of $f(x)$, it follows that

$$f(-x) = \sum_{n=1}^{\infty} a_n(-x)^n = \sum_{n=1}^{\infty} a_n(-1)^n x^n.$$

Consequently, if f is even, *i.e.*, $f(-x) = f(x)$, then

$$0 = f(x) - f(-x) = \sum_{n=1}^{\infty} a_nx^n - \sum_{n=1}^{\infty} a_n(-1)^n x^n = \sum_{n=1}^{\infty} a_n\big(1-(-1)^n\big)x^n$$

This series is identically zero, so each coefficient must vanish:

$$a_n\big((-1)^n - 1\big) = 0.$$

For even n, $(-1)^n - 1 = 1 - 1 = 0$, as required, but for odd n, $(-1)^n - 1 = -1 - 1 = -2$, which implies

$$a_n = 0 \quad \text{if } n \text{ is odd.}$$

Thus, if $f(x)$ is even, all coefficients a_n with odd n are zero.

Now suppose $f(x)$ is odd, *i.e.*, $f(-x) = -f(x)$. Then $f(-x) = -f(x)$, so

$$0 = f(x) + f(-x) = \sum_{n=1}^{\infty} a_nx^n - \sum_{n=1}^{\infty} a_n(-1)^n x^n = \sum_{n=1}^{\infty} a_n\big(1+(-1)^n\big)x^n$$

Again, each coefficient must vanish, so

$$a_n\big(1+(-1)^n\big) = 0.$$

For odd n, we have $1 + (-1)^n = 1 - 1 = 0$, but for even n, $(-1)^n + 1 = 2$, so we must have

$$a_n = 0 \quad \text{if } n \text{ is even.}$$

Hence, if $f(x)$ is odd, all coefficients a_n with even n are zero. $\square$